

Adora PINN — Step 3: Pretraining (warm start)

As-built notes · Ivan Milic · 2026-09-20 · step 3 of pinn/plan.md

Step 3 pretrains the step-1 field to the step-2 atmosphere so the PINN starts warm: a plain supervised fit of $f_\theta(x, y, z)$ to the stored cube (no RT yet). Files [pinn/pretrain.py](#), [pinn/step3_pretrain.ipynb](#); warm-start weights saved to [/dat/milic/adora_pinn_develop/pretrained_field.npz](#).

Method. Random minibatches (8192) of the $128 \times 128 \times 56$ grid points; a per-channel balanced loss — P compared in $\log_{10}$, the rest scaled ($T/10^3$, $v_z/3 \cdot 10^3$, $B/0.05$, $v_{\text{turb}}/10^3$); **optax** Adam, cosine LR 3×10^{-3} , 6000 steps (~ 3 min CPU, with a tqdm progress bar). Coordinates use **af.grid.coords**, which matches the cube's (x, y, z) flatten order.

Result. Loss $0.79 \rightarrow 1.0 \times 10^{-5}$. Full-grid RMS vs the target: T 3.96 K ($\sim 0.05\%$), P 4.1×10^{-3} dex ($\sim 0.9\%$), v_z 14 m/s (of ± 3 km/s), v_{turb} 4 m/s, $B \approx 0$. The field reproduces the FAL-C stratification and the granulation-like v_z pattern — a clean warm start. The notebook shows the loss curve, field-vs-target profiles ($T, \log P, v_{\text{turb}}, v_z$), and v_z target/field/residual maps.

Next. Step 4: synthesize the 6301/6302 spectrum from this initial model. Files uncommitted.